Weighted Adaptive Stimulation After Batch-testing Identification (WASABI) Problem Formulation

There are a few ways we can formulate the question

1. Similar to Draelos 2020 first identify the binary connections using group testing, then use the identified connection to fit a weighted model.
2. Directly use a linear-Poisson model to identify weights using group testing. (Works on the assumption that connections are linear)
3. If not linear, use NN or some other nonlinear approaches.

We will focus on (2) first, with fixed continuous stimulation parameters (e.g., current, frequency, pulse width).

Simplified formulation

$$\begin{aligned} \mathbf{r} &\sim \text{Poisson}(\boldsymbol{\lambda}) \\ \boldsymbol{\lambda}^{\text{stim}} &= \mathbf{C}\mathbf{s} + \boldsymbol{\lambda}^{\text{control}} \\ \mathbf{C} &= \mathbf{D}\mathbf{W}_{\text{low-rank}} \end{aligned}$$

$\mathbf{r}$: Recorded responses (spikes)
$\boldsymbol{\lambda}$: Firing rates
$\mathbf{C}$: Mapping from stimulation channels to firing rates
$\mathbf{s}$: Selection of stimulation channels
$\mathbf{D}$: Mapping from latent variables to recorded responses
$\mathbf{W}_{\text{low-rank}}$: Low rank mapping from stimulation channels to latent variables

- $\mathbf{W}$ is unknown.
- At time t-1, we want to find $\mathbf{s}_t$ that maximizes the information about $\mathbf{W}$, given $\mathbf{r}_{1:t-1}$ and $\mathbf{s}_{1:t-1}$

Ridge regression for learning W at time t

If we assume we know D_t at time t, we estimate $\mathbf{W}_t$ using a ridge least-squares objective:

$$\hat{\mathbf{W}}_t = \arg\min_{\mathbf{W}} ||\mathbf{Z} - \mathbf{WS}||_F^2 + \beta||\mathbf{W}||_F^2$$

where β is the weight of the ridge prior. We can expanding the Frobenius norm using trace:

$$\mathcal{L}(\hat{\mathbf{W}}_t) = tr[(\mathbf{Z} - \mathbf{WS})(\mathbf{Z} - \mathbf{WS})^\top] + \beta\, tr(\mathbf{WW}^\top)$$
$$\mathcal{L}(\hat{\mathbf{W}}_t) = tr(\mathbf{ZZ}^\top) - 2\, tr(\mathbf{ZS}^\top\mathbf{W}^\top) + tr(\mathbf{WSS}^\top\mathbf{W}^\top) + \beta\, tr(\mathbf{WW}^\top)$$

Keep terms relative to $\mathbf{W}_t$

$$\mathcal{L}(\hat{\mathbf{W}}_t) = -2\,\mathrm{tr}(\mathbf{ZS}^\top\mathbf{W}^\top) + \mathrm{tr}(\mathbf{WSS}^\top\mathbf{W}^\top) + \beta\,\mathrm{tr}(\mathbf{WW}^\top)$$

Taking derivative and set to zero:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{W}} &= -2\,\mathbf{ZS}^\top + 2\,\mathbf{WSS}^\top + 2\beta\,\mathbf{W} \\
0 &= -2\,\mathbf{ZS}^\top + 2\,\mathbf{WSS}^\top + 2\beta\,\mathbf{W} \\
\mathbf{WSS}^\top + \beta\mathbf{W} &= \mathbf{ZS}^\top \\
\mathbf{W}(\mathbf{SS}^\top + \beta\mathbf{I}) &= \mathbf{ZS}^\top
\end{aligned}$$

$$\boxed{\mathbf{W} = \mathbf{ZS}^\top(\mathbf{SS}^\top + \beta\mathbf{I})^{-1}}$$

Generative model for perturbation

$$(\boldsymbol{\lambda}_{\text{stim}} - \boldsymbol{\lambda}_{\text{control}}) = \mathbf{D}(\mathbf{WS} + \boldsymbol{\xi}) + \boldsymbol{\eta}$$

Note:

- Current loss is written for estimating $\mathbf{W}_t$
- $\mathbf{D}$ - learned from control? or learn jointly with trials t (Alternateing or SVD)?
- $\boldsymbol{\xi}_t$ is perturbation noise (e.g. mismatch between intended and actual stim-response effects) - hyperparameter, match data?
- $\boldsymbol{\eta}$: observation noise - hyperparameter, match data?
- Columns of $\mathbf{W}$ have unit variance
- Ridge prior should scale as $\beta \propto \frac{1}{n}$ so in low-trial regime solution is not degenerate and high trial regime data info is respected.

Minimizing the posterior covariance of W (Gaussian)

For adaptive selection of stimulation patterns, at each trial t, we want to select $\mathbf{s}_t$ that maximizes the expected information gain about $\mathbf{W}$. Equivalently, this would be selecting $\mathbf{s}_t$ that minimizes the determinant of the covariance of $\mathbf{W}$ at time t. We separate out the Poisson MLE rate estimation and identification of the optimal $\mathbf{W}^*$ in our formulation. This formulation could be good for control-theoretic implementations later too.

$$\boldsymbol{\lambda}_t = \mathbf{D}\mathbf{W}\mathbf{s}_t + \boldsymbol{\lambda}_t^{control}$$

Here, we are trying to find the optimal $\mathbf{W}^*$, where $\boldsymbol{\lambda}$, $\mathbf{s}_t$, and $\boldsymbol{\lambda}_t^{control}$ are known.

Let Σ_{t-1}	be the covariance matrix of $\mathbf{W}$ before trial $t-1$
Let β	be how we scale our data relative to our prior (could be $1/n$)

In terms of precision, we can formulate the Gaussian case as:

$$\Sigma_t^{-1} = \Sigma_{t-1}^{-1} + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T$$

Take the determinant

$$\det \Sigma_t^{-1} = \det(\Sigma_{t-1}^{-1} + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T)$$

using the matrix inversion lemma:

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^T) = (1 + \mathbf{v}^T\mathbf{A}^{-1}\mathbf{u})\det(\mathbf{A})$$

$$\det(\Sigma_{t-1}^{-1} + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T) = (1 + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^T\Sigma_{t-1}\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t])\det(\Sigma_{t-1}^{-1})$$

Since we minimize over $\mathbf{s}_t$, the objective function is therefore:

$$\begin{aligned}\arg\max_{\mathbf{s}}\ \det\left(\Sigma_t^{-1}\right) &= \arg\max_{\mathbf{s}}\ \left(1 + \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^\top\Sigma_{t-1}\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\right)\det(\Sigma_{t-1}^{-1}) \\ &= \arg\max_{\mathbf{s}}\ \beta\, \mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]^\top\Sigma_{t-1}\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t]\end{aligned}$$

We can further expand out:

$$\mathbb{E}[\boldsymbol{\lambda}_t \mid \mathbf{s}_t] = \hat{\mathbf{D}}_{t-1}\hat{\mathbf{W}}_{t-1}\mathbf{s}_t + \boldsymbol{\lambda}_t^{control}$$

where $\hat{\mathbf{D}}_{t-1}$ and $\hat{\mathbf{W}}_{t-1}$ are estimated from timepoints $i = 1, \ldots, t-1$ and $\boldsymbol{\lambda}_t^{control}$ is firing rate identified from control reaches.

Dynamics model

Instead of assuming an instantaneous mapping $\mathbf{W}$, we model stimulation as a perturbation to a latent dynamical system through the mapping $\mathbf{B}$. We can also just combine rate estimation with the observation equation (also dimensionality reduction)

State equation

$$\begin{aligned}
\mathbf{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\mathbf{x}_t &= \mathbf{A}\mathbf{x}_{t-1} + \mathbf{B}\mathbf{s}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \\
\mathbf{y}_t &\sim \text{Poisson}\big(\text{softplus}(\mathbf{C}\mathbf{x}_t + \mathbf{d})\big)
\end{aligned}$$

where:

- $\mathbf{A} \in \mathbb{R}^{K \times K}$ Dynamics operator
- $\mathbf{B} \in \mathbb{R}^{K \times N}$: Stimulus-to-latent mapping
- $\mathbf{C} \in \mathbb{R}^{N \times K}$: Observation (loading) matrix
- $\mathbf{D} \in \mathbb{R}^{N \times N}$: Direct ch-to-ch effect that is a diagonal matrix
- $\mathbf{d} \in \mathbb{R}^{N}$: Per-channel mean log firing rate
- $\mathbf{Q} \in \mathbb{R}^{K \times K}$: Latent process noise covariance
- $\mathbf{s}_t \in \mathbb{R}^{N}$: One-hot encoding of stimulus input at time t (Control trials just have $\vec{\mathbf{0}}$)
- $\mathbf{y}_t \in \mathbb{R}^{N}$: Binned spike counts
- $K \ll N$ Low-rank assumption

Fitting using EKF-style integral and SGD (Ignore Ds for now)

The marginal likelihood is:

$$\log p(\mathbf{y}_{1:t}|\boldsymbol{\theta}) = \log \int p(\mathbf{y}_{1:t} \mid \mathbf{x}_{1:t}; \boldsymbol{\theta})\, p(\mathbf{x}_{1:t}; \boldsymbol{\theta})\, d\mathbf{x}_{1:t}$$

$$\mathbf{y}_t \sim \text{Poisson}\big(\text{softplus}(\mathbf{C}\mathbf{x}_t + \mathbf{d})\big)$$

$$\boldsymbol{\theta} = \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{d}, \mu_{\mathbf{0}}, \boldsymbol{\Sigma}_{\mathbf{0}}\}$$

Because the Poisson emission $p(\mathbf{y}_{1:T} \mid \mathbf{x}_{1:T}; \boldsymbol{\theta})$ is non-Gaussian, the integral is intractable. Need to approximate it by linearizing the emission around $\hat{\mathbf{x}}_t$, the current filtered mean (latent states), yielding a Gaussian approximation to the predictive distribution

$$p(\mathbf{y}_t \mid \mathbf{y}_{1:t-1}; \boldsymbol{\theta}) \approx \mathcal{N}(\boldsymbol{\lambda}_{t|t-1},\, \mathbf{S}_t)$$

where

$$\boldsymbol{\lambda}_{t|t-1} = \text{softplus}\Big(\mathbf{C}\hat{\mathbf{x}}_{t|t-1} + \mathbf{d}\Big)$$

and $\mathbf{S}_t$ is the estimated covariance.

This gives the approximate marginal likelihood

$$\mathcal{L}(\boldsymbol{\theta}) \approx \sum_{t=1}^{T} \log \mathcal{N}(\mathbf{y}_t |\, \boldsymbol{\lambda}_{t|t-1},\, \mathbf{S}_t)$$

which we can maximize directly via SGD.

Emission Jacobian For $\boldsymbol{\lambda}_t = \text{softplus}(\mathbf{C}\hat{\mathbf{x}}_t + \mathbf{d})$, we have the Jacobian (log-rate with respect to the latent state) is:

$$\mathbf{H}_t = \left.\frac{\partial \boldsymbol{\lambda}_t}{\partial \mathbf{x}_t}\right|_{\hat{\mathbf{x}}_t} = \text{diag}(\sigma(\mathbf{C}\hat{\mathbf{x}}_t + \mathbf{d}))\, \mathbf{C} \in \mathbb{R}^{N \times K}$$

where $\sigma(\cdot)$ is the sigmoid function (derivative of softplus). We calculate the Emission Jacobian $\mathbf{H}_t$ to linearize the non-Gaussian Poisson emission and plug into the Kalman equations. This propagates the Gaussian estimations of latent states $\mathbf{x}_t$ one time step forward each.

EKF forward pass prediction update

$$
\begin{aligned}
\hat{\mathbf{x}}_{t|t-1} &= \mathbf{A}\hat{\mathbf{x}}_{t-1|t-1} + \mathbf{B}\mathbf{s}_t \\
\mathbf{P}_{t|t-1} &= \mathbf{A}\mathbf{P}_{t-1|t-1}\mathbf{A}^\top + \mathbf{Q} \\
\mathbf{S}_t &= \mathbf{H}_t\mathbf{P}_{t|t-1}\mathbf{H}_t^\top + \mathrm{diag}(\boldsymbol{\lambda}_{t|t-1})^{-1} \\
\mathbf{K}_t &= \mathbf{P}_{t|t-1}\mathbf{H}_t^\top\mathbf{S}_t^{-1} \\
\hat{\mathbf{x}}_{t|t} &= \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t(\mathbf{y}_t - \boldsymbol{\lambda}_{t|t-1}) \\
\mathbf{P}_{t|t} &= (\mathbf{I} - \mathbf{K}_t\mathbf{H}_t)\mathbf{P}_{t|t-1}
\end{aligned}
$$

where

- $\hat{\mathbf{x}}_{t|t-1} \in \mathbb{R}^K$: predicted latent state mean
- $\hat{\mathbf{x}}_{t|t} \in \mathbb{R}^K$: filtered latent state mean
- $\mathbf{P}_{t|t-1} \in \mathbb{R}^{K\times K}$: predicted latent state covariance
- $\mathbf{P}_{t|t} \in \mathbb{R}^{K\times K}$: filtered latent state covariance
- $\mathbf{S}_t \in \mathbb{R}^{N\times N}$: innovation covariance
- $\mathbf{K}_t \in \mathbb{R}^{K\times N}$: Kalman gain
- $(\mathbf{y}_t - \boldsymbol{\lambda}_{t|t-1})$: innovation (residual between observed spikes and predicted rate)
- $\mathbf{Q} \in \mathbb{R}^{K\times K}$: latent process noise covariance
- $\mathrm{diag}(\boldsymbol{\lambda}_{t|t-1})^{-1}$: Poisson observation noise (variance equals mean)
- $\mathbf{H}$: locally linear estimation of the emission matrix

EKF marginal log-likelihood With this we have the defined objective with the approximated $\boldsymbol{\lambda}_{t|t-1}, \mathbf{S}_t$

$$
\mathcal{L}(\boldsymbol{\theta}) \approx \sum_{t=1}^{T} \log \mathcal{N}(\mathbf{y}_t |\, \boldsymbol{\lambda}_{t|t-1},\, \mathbf{S}_t)
$$

SGD parameter update Parameters are updated by differentiating through the EKF pass:

$$
\boldsymbol{\theta}^{(\tau+1)} = \boldsymbol{\theta}^{(\tau)} - \eta_\tau \, \nabla_{\boldsymbol{\theta}} \left[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\right]
$$

where η_τ is the effective learning rate defined by the Adam optimizer. More specifically:

$$
\begin{aligned}
\mathbf{m}_\tau &= \beta_1 \mathbf{m}_{\tau-1} + (1-\beta_1)\, \nabla_{\boldsymbol{\theta}}\big[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\big] && \text{(momentum)} \\
\mathbf{v}_\tau &= \beta_2 \mathbf{v}_{\tau-1} + (1-\beta_2)\, \nabla_{\boldsymbol{\theta}}\big[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\big]^2 && \text{(RMSprop)} \\
\hat{\mathbf{m}}_\tau &= \frac{\mathbf{m}_\tau}{1-\beta_1^\tau}, \qquad \hat{\mathbf{v}}_\tau = \frac{\mathbf{v}_\tau}{1-\beta_2^\tau} && \text{(bias correction)} \\
\boldsymbol{\theta}^{(\tau+1)} &= \boldsymbol{\theta}^{(\tau)} - \frac{\alpha}{\sqrt{\hat{\mathbf{v}}_\tau} + \epsilon}\, \hat{\mathbf{m}}_\tau && \text{(parameter update)}
\end{aligned}
$$

where $\mathbf{m}_\tau$ and $\mathbf{v}_\tau$ are the first and second moment estimates, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ are the default hyperparameters, and α is the base learning rate.

and gradients are computed via automatic differentiation in JAX in the dynamax package. The full parameter set is:

$$
\boldsymbol{\theta} = \{\mathbf{A},\, \mathbf{B},\, \mathbf{C},\, \mathbf{d},\, \mathbf{Q},\, \boldsymbol{\mu}_0,\, \boldsymbol{\Sigma}_0\}
$$

Information criterion for B, minimizing the posterior covariance of B (Poisson LDS)

Goal: Estimate uncertainty of $\mathbf{B}$
Posterior $p(\mathbf{B} \mid \mathbf{y}_{1:t}, \mathbf{s}_t)$

$$\begin{aligned}
\mathbf{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\mathbf{x}_t &= \mathbf{A}\mathbf{x}_{t-1} + \mathbf{B}\mathbf{s}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \\
\mathbf{y}_t &\sim \text{Poisson}\big(\text{softplus}(\mathbf{C}\mathbf{x}_t + \mathbf{d})\big)
\end{aligned}$$

Likelihood after marginalizing out $\mathbf{x}_{1:t}$:

$$p(\mathbf{y}_{1:t} \mid \boldsymbol{\theta}) = \int \underbrace{p(\mathbf{y}_{1:t} \mid \mathbf{x}_{1:t}, \boldsymbol{\theta})}_{\text{Poisson emission, need EKF approx.}} \underbrace{p(\mathbf{x}_{1:t} \mid \mathbf{s}_t, \boldsymbol{\theta})}_{\text{Gaussian dynamics prior}} d\mathbf{x}_{1:t}$$

$$\boldsymbol{\theta} = \{\mathbf{A},\, \mathbf{B},\, \mathbf{C},\, \mathbf{d},\, \mathbf{Q},\, \boldsymbol{\mu}_0,\, \boldsymbol{\Sigma}_0\}$$

Bayes':

$$\underbrace{p(\boldsymbol{\theta} \mid \mathbf{y}_{1:t})}_{\text{Posterior}} = \frac{\overbrace{p(\mathbf{y}_{1:t} \mid \boldsymbol{\theta})}^{\text{likelihood}} \cdot \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\mathbf{y}_{1:t})}_{\text{evidence, constant in } \boldsymbol{\theta}}}$$

Assume $\boldsymbol{\theta} \setminus \mathbf{B} = \mathbf{A}, \mathbf{C}, \mathbf{d}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0$ known from Poisson LDS fit via EKF SGD, and $\mathbf{s}_t$ known.

$$\begin{aligned}
p(\boldsymbol{\theta} \mid \mathbf{y}_{1:t}) &\propto p(\mathbf{y}_{1:t} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}) \\
p(\mathbf{B} \mid \mathbf{y}_{1:t}, \mathbf{s}_t) &\propto p(\mathbf{y}_{1:t} \mid \mathbf{B},\, \boldsymbol{\theta} \setminus \mathbf{B}) \cdot p(\mathbf{B}) \cdot p(\mathbf{A}) \cdot p(\mathbf{C}) \cdots \quad \text{(by independence of } \boldsymbol{\theta}\text{)}
\end{aligned}$$

Absorbing $p(\boldsymbol{\theta} \setminus \mathbf{B})$ into the constant and expanding over $\mathbf{x}_{1:t}$:

$$p(\mathbf{B} \mid \mathbf{y}_{1:t}, \mathbf{s}_t) \propto \underbrace{p(\mathbf{B})}_{\text{Initialize via } \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_0^B)} \int \underbrace{p(\mathbf{y}_{1:t} \mid \mathbf{x}_{1:t}, \mathbf{B}, \boldsymbol{\theta} \setminus \mathbf{B})}_{\text{Poisson emission}} \underbrace{p(\mathbf{x}_{1:t} \mid \mathbf{s}_t, \boldsymbol{\theta})}_{\text{depends on } \mathbf{B}\text{,can get from EKF}} d\mathbf{x}_{1:t} \tag{1}$$

Need EKF to estimate. TODO: Make sure we can split up theta components in dynamax.

One step ahead prediction in latent space to get one step ahead $\mathrm{Cov}(\hat{\mathbf{B}}_{t|t-1})$ estimation

At time step $t-1$:

$$\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \mathbf{B}\mathbf{s}_t + \boldsymbol{\epsilon}_t$$

Rearranging to isolate $\hat{\mathbf{B}}_{t|t-1}\mathbf{s}_t$, and substituting EKF filtered means AND predicted, poststim means $\hat{\mathbf{x}}_{t-1|t-1}, \hat{\mathbf{x}}_{t|t-1,\text{stim}}$, in place of true latent states, and accurate description of estimations:

$$\hat{\mathbf{B}}_{t|t-1}\mathbf{s}_t = \hat{\mathbf{x}}_{t|t-1,\text{stim}} - \hat{\mathbf{A}}\hat{\mathbf{x}}_{t-1|t-1} - \boldsymbol{\epsilon}_t$$

where:

- $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$
- $\hat{\mathbf{x}}_{t-1|t-1} \sim \mathcal{N}(\mathbf{x}_{t-1}, \mathbf{P}_{t-1|t-1})$
- $\hat{\mathbf{x}}_{t|t-1,\text{stim}} = \hat{\mathbf{A}}\hat{\mathbf{x}}_{t-1|t-1} + \hat{\mathbf{B}}_{t-1|t-1}\mathbf{s}_t + \boldsymbol{\epsilon}_t$
- For now we ignore $\mathrm{Cov}(\hat{\mathbf{A}})$

Now we want the covariance:

$$\begin{aligned}
\boldsymbol{\Sigma}_{B_{t|t-1}} = \mathrm{Cov}(\hat{\mathbf{B}}_{t|t-1}) &= \mathrm{Cov}(\hat{\mathbf{x}}_{t|t-1,\text{stim}}) + \mathrm{Cov}(\hat{\mathbf{A}}\hat{\mathbf{x}}_{t-1|t-1}) + \mathrm{Cov}(\boldsymbol{\epsilon}_t) \\
\boldsymbol{\Sigma}_{B_{t|t-1}} &\approx \mathrm{Cov}(\hat{\mathbf{A}}\hat{\mathbf{x}}_{t-1|t-1} + \hat{\mathbf{B}}_{t-1|t-1}\mathbf{s}_t + \boldsymbol{\epsilon}_t) + \hat{\mathbf{A}}\mathbf{P}_{t-1|t-1}\hat{\mathbf{A}}^\top + \mathbf{Q} \\
\boldsymbol{\Sigma}_{B_{t|t-1}} &\approx \mathrm{Cov}(\hat{\mathbf{B}}_{t-1|t-1}\mathbf{s}_t) + 2\hat{\mathbf{A}}\mathbf{P}_{t-1|t-1}\hat{\mathbf{A}}^\top + 2\mathbf{Q}
\end{aligned}$$

Let $\hat{\mathbf{B}}_{t-1|t-1}\mathbf{s}_t$ be $\mathbf{B}\mathbf{s}$ for now.

$$\mathrm{Cov}(\mathbf{B}\mathbf{s}) = \mathbb{E}\Big[\mathbf{B}\mathbf{s}\mathbf{s}^T\mathbf{B}^T\Big] - \mathbb{E}[\mathbf{B}\mathbf{s}]\,\mathbb{E}[\mathbf{B}\mathbf{s}]^T$$

IID case, which is wrong.

$$\begin{aligned}
\mathbf{B}_{ij} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \\
\mathrm{Cov}(\mathbf{B}\mathbf{s}) &= \mathbb{E}\left[\sum_{j\in\mathbf{s}_j} \mathbf{B}_{ij} \sum_{j\in\mathbf{s}_j} \mathbf{B}_{kj}\right] - \mathbb{E}\left[\sum_{j\in\mathbf{s}_j} \mathbf{B}_{ij}\right] \mathbb{E}\left[\sum_{j\in\mathbf{s}_j} \mathbf{B}_{kj}\right] \\
\mathrm{Cov}(\mathbf{B}\mathbf{s}) &= \begin{cases} \sum_{j\in\mathbf{s}_j} \mathrm{Var}(\mathbf{B}_{ij}) & \text{if } i = k \\ 0 & \text{else} \end{cases} \\
\mathrm{Cov}(\hat{\mathbf{B}}_{t-1|t-1}\,\mathbf{s}_t) &= \mathrm{diag}\big(\boldsymbol{\Sigma}_{t-1,B}\,\mathbf{s}_t\big)\,, \text{since } \mathbf{s}_t \text{ is a one-hot encoding vector.}
\end{aligned}$$

But more accurately, to capture the dependencies across channels (not IID), use the posterior of $\mathbf{B}$ from (1). Since $\mathbb{E}[\mathbf{B}] = \hat{\mathbf{B}}_{t-1|t-1}$:

$$\begin{aligned}
\mathrm{Cov}(\mathbf{B}\mathbf{s}) &= \mathbb{E}\Big[\mathbf{B}\mathbf{s}\mathbf{s}^T\mathbf{B}^T\Big] - \mathbb{E}[\mathbf{B}\mathbf{s}]\,\mathbb{E}[\mathbf{B}\mathbf{s}]^T \\
\mathrm{Cov}(\mathbf{B}\mathbf{s}) &= \mathbb{E}[\mathbf{B}]\,\mathbf{s}\mathbf{s}^T\mathbb{E}\Big[\mathbf{B}^T\Big] + \mathrm{Cov}(\mathbf{B})\cdot\mathbf{s}\mathbf{s}^T - \mathbb{E}[\mathbf{B}]\mathbf{s}\mathbf{s}^T\mathbb{E}[\mathbf{B}]^T \\
\mathrm{Cov}(\hat{\mathbf{B}}_{t-1|t-1}\,\mathbf{s}_t) &= \boldsymbol{\Sigma}_{B_{t-1|t-1}}\,\mathbf{s}_t\mathbf{s}_t^\top
\end{aligned}$$

Substituting back:

$$\boldsymbol{\Sigma}_{B_{t|t-1}} \approx \boldsymbol{\Sigma}_{B_{t-1|t-1}}\,\mathbf{s}_t\mathbf{s}_t^\top + 2\hat{\mathbf{A}}\mathbf{P}_{t-1|t-1}\hat{\mathbf{A}}^\top + 2\mathbf{Q}$$

How do we initialize to make sure $\boldsymbol{\Sigma}_{B_0}$ is properly initialized when a given $\mathbf{s}_t$ hasn't been tested?

Acquisition function for B (Poisson LDS)

For adaptive selection of stimulation patterns, at each trial t, we want to select $\mathbf{s}_t$ that maximizes the expected information gain about $\mathbf{B}$. Equivalently, this would be selecting $\mathbf{s}_t$ that minimizes the determinant of the covariance of $\mathbf{B}$ at time t.

Let $\mathbf{\Sigma}_{t-1,B}^{-1}$	be the precision matrix of $\mathbf{B}$ before trial $t-1$
Let $\mathbf{\Delta}_t = \mathbf{B}\mathbf{s}_t$	be the latent perturbation induced by stimulation pattern $\mathbf{s}_t$
Let β	be how we scale our data relative to our prior (could be $1/n$)

In terms of precision, we can formulate the Gaussian case as:

$$\mathbf{\Sigma}_{t,B}^{-1} = \mathbf{\Sigma}_{t-1,B}^{-1} + \beta\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]\mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]^\top$$

Taking the determinant and applying the matrix determinant lemma:

$$\begin{aligned}\det \mathbf{\Sigma}_t^{-1,B} &= \det\Big(\mathbf{\Sigma}_{t-1,B}^{-1} + \beta\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]\mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]^\top\Big) \\ &= \Big(1 + \beta\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]^\top \mathbf{\Sigma}_{t-1}^{B}\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]\Big) \det\big(\mathbf{\Sigma}_{t-1,B}^{-1}\big)\end{aligned}$$

Since we minimize over $\mathbf{s}_t$, the objective function is therefore:

$$\begin{aligned}\arg\max_{\mathbf{s}}\ \det\left(\mathbf{\Sigma}_{t,B}^{-1}\right) &= \arg\max_{\mathbf{s}}\ \Big(1 + \beta\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]^\top \mathbf{\Sigma}_{t-1}^{B}\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]\Big) \det\big(\mathbf{\Sigma}_{t-1,B}^{-1}\big) \\ &= \arg\max_{\mathbf{s}}\ \beta\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]^\top \mathbf{\Sigma}_{t-1}^{B}\, \mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t]\end{aligned}$$

We can further expand out the expected perturbation in latent space:

$$\mathbb{E}[\mathbf{\Delta}_t \mid \mathbf{s}_t] = \hat{\mathbf{B}}_{t-1}\mathbf{s}_t$$

where $\hat{\mathbf{B}}_{t-1}$ is estimated from timepoints $i = 1, \ldots, t-1$, giving the final objective:

$$\arg\max_{\mathbf{s}}\ \beta\, \mathbf{s}_t^\top \hat{\mathbf{B}}_{t-1}^\top \mathbf{\Sigma}_{t-1}^{B}\, \hat{\mathbf{B}}_{t-1}\mathbf{s}_t$$

Problems:

1. Need to expand to KxN for the precision matrix, instead of a K x K matrix. Current formulation gives you which latent to stim, which gives you probably channels just based on its contribution/composition of the latents
2. Compositional model?
3. This is for one timepoint, don't worry too much about some kind of t to t+k information update, before deciding on the next step, where t+k is the next stim
4. Tried fitting the PLDS with a step term ($\mathbf{C} \times \text{step}(t)$) before, but doesn't seem to do well.
5. Try to find closed form after plugging in $\mathbb{E}[\mathbf{B}]$?

Dynamics model + Behavior (Not rigorous, for class project)

$$
\begin{aligned}
\mathbf{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\mathbf{x}_t &= \mathbf{A}\mathbf{x}_{t-1} + \mathbf{B}\mathbf{s}_t + \mathbf{F}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \\
\mathbf{y}_t &\sim \text{Poisson}\big(\text{softplus}(\mathbf{C}\mathbf{x}_t + \mathbf{d} + \mathbf{D}\mathbf{s}_t)\big) \\
\mathbf{z}_t &\sim \mathcal{N}(\mathbf{G}\mathbf{x}_t + \mathbf{g}, \mathbf{R})
\end{aligned}
$$

where:

- $\mathbf{A} \in \mathbb{R}^{K \times K}$: Dynamics operator
- $\mathbf{B} \in \mathbb{R}^{K \times N}$: Stimulus-to-latent mapping
- $\mathbf{C} \in \mathbb{R}^{N \times K}$: Observation (loading) matrix
- $\mathbf{D} \in \mathbb{R}^{N \times N}$: Direct ch-to-ch effect, diagonal matrix
- $\mathbf{d} \in \mathbb{R}^{N}$: Per-channel mean log firing rate
- $\mathbf{F} \in \mathbb{R}^{K \times 12}$: Kinematics-to-latent feedback mapping
- $\mathbf{G} \in \mathbb{R}^{12 \times K}$: Latent-to-kinematics readout matrix
- $\mathbf{g} \in \mathbb{R}^{12}$: Per-channel kinematic bias
- $\mathbf{Q} \in \mathbb{R}^{K \times K}$: Latent process noise covariance
- $\mathbf{R} \in \mathbb{R}^{12 \times 12}$: Kinematic observation noise covariance
- $\mathbf{s}_t \in \mathbb{R}^{N}$: One-hot encoding of stimulus input at time t (control trials have $\mathbf{s}_t = \mathbf{0}$)
- $\mathbf{y}_t \in \mathbb{R}^{N}$: Binned spike counts
- $\mathbf{z}_t \in \mathbb{R}^{12}$: Stacked IMU position and velocity from wrist and upper arm bands, $\mathbf{z}_t = [\mathbf{p}_{w,t}^\top,\ \mathbf{v}_{w,t}^\top,\ \mathbf{p}_{u,t}^\top,\ \mathbf{v}_{u,t}^\top]^\top$
- $K \ll N$: Low-rank assumption

TODO

- Since s is Bernoulli (discrete), projected or coordinate or proximal descent to solve for $\mathbf{s}$
- Forgot to index via trial in Poisson MLE section?
- Projected is projecting the final to discrete, proximal is projecting each step
- Simulate the Linear-Poisson model to make sure statistics align

Random Sampling Stopgap

- Total configurations: N
- Take N samples: stimulate groups of 2 sites (one row).
- For 128 sites, this is 64 rows.
- Each stim randomly sample 20%: 12 rows (can revisit this number)
- Collect all the data and responses and see if we can predict in a leave-one-out approach (use all but one response).
- Benefit of random sampling is that we can collect now before Nike's implant comes out.